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A Conceptual Framework for Reuse-Driven Moisture Sensing and Data Acquisition in Mass-Timber Buildings

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ABSTRACT

Moisture exposure is a major durability risk in mass-timber buildings, affecting service life, structural reliability, maintenance needs, and future reuse potential. Although embedded moisture sensors are increasingly used during construction and operation, their data are commonly managed for immediate detection, control, or maintenance rather than as lifecycle evidence for durability and reuse assessment. Guidance also remains limited on translating moisture-vulnerable building locations into risk-based sensor deployment that supports later interpretation. This paper proposes a conceptual framework for lifecycle-oriented moisture monitoring in mass-timber buildings. Drawing on research on timber moisture monitoring, durability, structural health monitoring, service-life assessment, maintenance planning, and lifecycle-oriented decision-making, it links monitoring objectives, moisture-risk classification, sensor strategy, contextual data requirements, exposure interpretation, and lifecycle data integration. The framework shows how moisture-critical zones can be translated into monitoring requirements, with location risk informing sensor deployment. It also outlines how raw sensor readings can be interpreted as indicators of moisture exposure, drying response, anomalies, and data quality. These records are intended to support maintenance, post-event assessment, and end-of-life evaluation, while positioning moisture data as one input to broader reuse assessment.

1. INTRODUCTION

Mass-timber construction is increasingly discussed as part of the transition towards lower-carbon and more circular building practices. Engineered wood products (EWPs) such as cross-laminated timber (CLT) and glued-laminated timber (glulam) can contribute to reduced embodied impacts when they replace more carbon-intensive structural materials, while also storing biogenic carbon during their service life (Balasbaneh & Sher, 2024). These benefits depend on design choices, production systems, transport, life-cycle assessment assumptions, and end-of-life scenarios (Duan et al., 2025). Extending the service life of timber elements through maintenance, repair, refurbishment, and reuse can therefore strengthen the environmental value of mass-timber construction.

Reuse of load-bearing timber products requires confidence in both the condition and history of the recovered element (Sabaa et al., 2026). For EWPs, this confidence depends not only on original product documentation and structural capacity, but also on evidence of service exposure, degradation risks, and damage arising during use or deconstruction. Timber's hygroscopic behaviour makes elevated moisture content (MC) a critical durability concern, as it can promote mould and fungal decay, dimensional instability, and reductions in structural reliability (Brischke & Rapp, 2008). Moisture

records could therefore provide useful evidence about exposure history, especially for concealed, inaccessible, or moisture-prone elements.

Moisture monitoring is used during construction and operation of mass-timber buildings, with existing studies providing practical evidence on sensor installation, measurement depth, and in-service data collection (Flexeder et al., 2022; Johns et al., 2024). In parallel, circular construction research increasingly emphasises traceability, documentation, and reliable condition information to support decisions beyond the initial use phase (Bucher et al., 2024). However, moisture monitoring data are not yet consistently structured or integrated as long-term, element-level records that can support maintenance, post-event assessment, or future reuse-related decisions (Sabaa et al., 2026). This points to a need for closer alignment between moisture monitoring practice and the information requirements of lifecycle assessment, durability management, and circular reuse.

This paper develops a conceptual framework for planning moisture monitoring in mass-timber buildings as a lifecycle information process. It emphasises how monitoring data can be structured, linked to element- and sensor-level information, and integrated into digital systems for lifecycle decision support. To develop this framework, the paper addresses two questions:

1. How can moisture-critical zones in mass-timber buildings be identified, classified, and translated into risk-based sensor deployment decisions?
2. What supporting data and contextual information, such as element identity, sensor location, and measurement depth, are needed to interpret moisture measurements for maintenance planning and end-of-life evaluation?

2. METHODOLOGY

This study developed a conceptual framework using conceptual framework analysis, following Jabareen's (2009) approach of identifying, categorising, integrating, and synthesising concepts from literature. This approach is suitable when knowledge is distributed across several fields and must be organised into an explanatory or prescriptive structure. The synthesis examined moisture-risk locations in timber buildings, how monitoring strategies can respond to those risks, and what information is needed to interpret moisture exposure in support of durability, maintenance, post-event, and future reuse-related assessment.

The framework was developed through a targeted literature synthesis combining database searching, snowballing, and review of relevant monitoring and timber-moisture studies. Titles, abstracts, and subsequently full texts were screened against the stated criteria. A total of 54 articles were assessed for relevance to timber moisture risk, monitoring practice, sensor deployment, moisture-exposure evaluation, and lifecycle use of monitoring records. Twenty articles were retained because they provided extractable evidence on sensor location, deployment configuration, contextual data requirements, or interpretation of monitoring records for maintenance or reuse-related decisions.

The retained studies were coded in NVivo. Inductively developed codes covered lifecycle stage, monitoring purpose, building and member type, sensor location, risk drivers, sensor configuration and density, installation guidance, monitoring issues, risk prediction, contextual information, and recommended monitoring systems. The codes supported systematic comparison across sources.

Framework development proceeded through iterative synthesis of the coded material. Coded excerpts were reviewed to distinguish evidence reported directly in sources from synthesis judgements developed through cross-source comparison. Concepts were grouped by their role in monitoring: moisture-risk conditions, monitoring locations, sensor deployment, contextual data, data quality, and exposure evaluation. These groupings were compared with coded decision factors:

exposure, drying capacity, inspection access, consequence, environmental aggressiveness, sensor survivability, and decision relevance. This comparison informed the framework structure and its risk-classification logic.

Exploratory analysis of two existing moisture monitoring datasets was also used to refine the framework (Baas et al., 2021; Riggio et al., 2019). Time-series records were examined descriptively through threshold-based identification of moisture events, comparison of moisture behaviour across monitored locations, and visual comparison with relative humidity, temperature, and relevant external exposure conditions. These observations were used to check whether the framework accounted for practical issues identified in the literature synthesis, including sensor reliability, environmental context, spatial variation, location-specific wetting and drying behaviour, and data-validation needs.

3. RESULTS

The analysis resulted in a lifecycle-oriented framework for moisture monitoring in mass-timber buildings (Fig. 1). Its eight linked layers connect why monitoring is needed, where sensors should be placed, how records should be interpreted, and how the resulting information can support decisions during construction, operation, and future assessment. To support application of the framework, a supporting repository is available at https://github.com/sosabaa/timber-moisture-framework; it provides a framework implementation example for a CLT kitchen-floor scenario, together with templates for risk-location classification, sensor deployment, and monitoring-record schema development.

3.1 Lifecycle context and moisture risk mapping — Layers 1 and 2

Layer 1 establishes when monitoring is needed and the intended use of the monitoring record before sensor placement is considered. This follows structural health monitoring (SHM) logic, in which monitoring objectives precede sensor deployment, data acquisition, interpretation, and response (Farrar & Worden, 2012). The lifecycle stage and intended use of the record together specify the monitoring context. Monitoring may be intended to support construction drying, enclosure approval, leak detection, maintenance planning, post-event assessment, repair or refurbishment, deconstruction, or future reuse assessment. Specifying this context helps align sensor deployment with the information required.

Layer 2 identifies and classifies moisture-risk locations. The literature shows that moisture performance in mass-timber buildings is strongly influenced by location, detail design, exposure condition, connection type, orientation, and material depth (Johns & Richman, 2025; Kalbe et al., 2020; Kellgren et al., 2023; Schmidt & Riggio, 2019). Moisture-risk mapping is therefore organised across building zones, assemblies and details, and element-level features where moisture may enter, accumulate, remain concealed, or cause damage. Examples include wet rooms, chemically aggressive interiors, areas exposed before enclosure, roof and façade interfaces, condensation-prone cavities, service penetrations, concealed connections, end-grain regions, fastening pockets, and interfaces with limited drying capacity (Johns & Richman, 2025; Olsson, 2021; Schmidt & Riggio, 2019; Vattenskadecentrum, 2025).

The classification criteria used in Layer 2 represent the authors' synthesis of the reviewed evidence. Candidate locations are assessed in terms of moisture risk and monitoring reliability, using exposure likelihood, drying capacity, consequence, inspection access, environmental aggressiveness, sensor survivability, and decision importance. This distinguishes locations with similar exposure but different monitoring needs, such as an accessible low-consequence surface compared with a concealed CLT roof-edge detail with limited drying capacity (Austigard & Mattsson, 2020; Schmidt & Riggio, 2019).

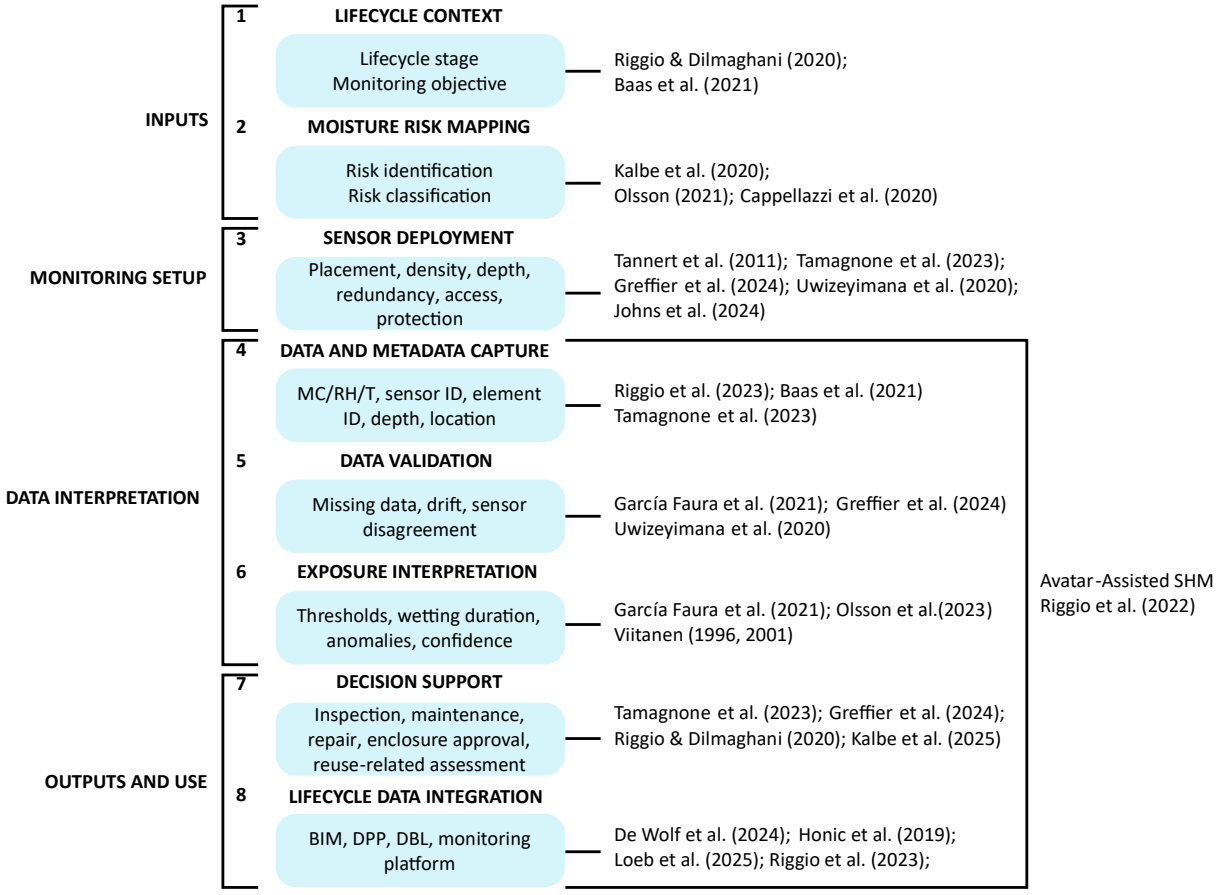


Figure 1: Conceptual framework for reuse-informed moisture sensing and data acquisition in mass-timber buildings. Extended layer-to-source mapping and supporting references are provided in the accompanying repository: https://github.com/sosabaa/timber-moisture-framework/tree/main/05_framework_evidence_base

3.2 Sensor deployment — Layer 3

Layer 3 translates the Layer 2 risk classification into decisions on sensor placement, density, measurement depth, redundancy, and protection (Fig. 1). Low-risk locations may be covered by reference measurements or periodic inspection; medium-risk locations by one representative sensor group; and high-risk locations by several sensors at vulnerable points. Critical locations, where data may inform enclosure approval, repair, warranty, or reuse-related assessment, may require redundancy, depth variation, and independent validation points.

Sensor placement should reflect the anticipated moisture mechanism. Shallow measurements are appropriate for surface wetting, condensation, or mould risk near exposed surfaces or beneath finishes. Deeper measurements are needed where moisture may penetrate or remain trapped within thick sections, behind membranes, around concealed connections, near beam ends, or at panel edges. Surface moisture may respond rapidly to wetting and drying, whereas internal moisture can persist and may be more relevant to durability assessment (Schmidt & Riggio, 2019).

Sensors at reference locations can support interpretation by distinguishing local wetting or delayed drying at vulnerable details from broader seasonal or indoor-climate behaviour. In the Peavy Hall dataset, an edge-by-hardware location reached approximately 27.6% MC and remained above 20% MC for about 88 days, whereas a lower-shearwall middle-panel location remained below 16% MC (Baas et al., 2021; Riggio et al., 2019).

Redundancy is appropriate where a single reading would provide insufficient confidence for the intended use of the data. It may include duplicate MC measurements at the same location, paired shallow and deep measurements, nearby reference sensors, RH and temperature measurements, or periodic manual verification. It is particularly relevant in inaccessible, high-consequence, aggressive, or non-replaceable locations, where sensor damage, drift, communication loss, or local material variability may affect interpretation (Tingstveit et al., 2020; Uwizeyimana et al., 2020).

Installation and protection should also reflect the identified risk conditions. Construction-stage installations may require protected cable routes, physical shielding, and documentation of installation conditions, particularly where wet CLT may later be covered by layers that restrict drying (Austigard & Mattsson, 2020). Sensors in wet rooms, façade cavities, roof or floor build-ups, and aggressive environments may require moisture-resistant housings, corrosion-resistant components, accessible junction boxes, or replaceable arrangements where feasible (Greffier et al., 2024). For each sensor group, the record should retain sensor position, measurement depth, installation conditions, protection measures, and any subsequent replacement or recalibration so that deployment choices remain interpretable in later use.

3.3 Data, validation and exposure interpretation — Layers 4, 5 and 6

The third component covers Layers 4–6 of the framework: data and metadata capture, validation, and exposure interpretation. Together, these layers describe how sensor readings become interpretable moisture-exposure records. The framework distinguishes between measured values, contextual information, and interpreted indicators.

Layer 4 captures the measured values and contextual information needed to preserve record meaning. Measured values include MC as the primary variable, or the raw electrical outputs from which it is derived, together with RH, temperature, timestamp, sensor identity, and measurement depth. Interpretation also requires element identity, assembly location, sensor type, calibration basis, correction method, installation date, sampling interval, risk class, exposure events, maintenance actions, and any replacement or recalibration. These metadata link the reading, physical timber element, and decision context (Baas et al., 2021; Riggio et al., 2022; Uwizeyimana et al., 2020). Table 1 summarises the information fields used in the framework.

Table 1: Core information requirements for interpretable moisture monitoring records

Information category	Examples
Measured variables	MC (primary variable) or raw sensor output from which MC is derived, supporting RH and temperature, timestamp
Sensor metadata	Sensor ID, sensor type, measurement depth, calibration basis, correction method, sampling interval
Element and location	Element ID, BIM element ID, material type, assembly location, sensor position, panel or junction reference
Risk and context information	Risk class, monitoring objective, exposure context, construction or service events
Validation and quality information	Data gaps, implausible values, drift, sensor damage, redundancy disagreement, incomplete metadata
Exposure indicators	Threshold exceedance, wetting duration, drying behaviour, anomalies, confidence or uncertainty
Lifecycle record information	Maintenance actions, inspections, interventions, repair history, links to future assessment records

Semantic data models provide a way to describe relationships between sensors, building elements, locations, measurements, metadata, and interpretation rules (Costa & Sicilia, 2022). In moisture monitoring, MC, RH, and T values remain meaningful when linked to element identity, sensor position, depth, installation context, risk class, and data quality. The avatar-assisted SHM concept proposed by Riggio et al. (2022) provides a related timber-monitoring example in which software entities process sensor data with contextual information, address data gaps, support validation, and produce decision-oriented outputs. Here, these ideas define the metadata, quality flags, and exposure indicators needed to preserve lifecycle meaning.

The record created in Layer 4 then passes to Layer 5: validation, before it is used for exposure interpretation. Long-term records may contain missing data, duplicate timestamps, implausible values, abrupt jumps, sensor drift, battery failure, communication gaps, sensor damage, corrosion, or disagreement between redundant sensors. The framework therefore includes quality flags for data gaps, implausible readings, sensor drift, communication problems, sensor damage, redundancy disagreement, and incomplete metadata. This follows SHM logic in which sensor data require validation, comparison, enrichment, and reasoning before decision support (Riggio et al., 2022).

Validated readings are carried forward to Layer 6: exposure interpretation. Five categories are used: threshold exceedance, moisture dose and duration, drying behaviour, anomaly detection, and confidence or uncertainty. These indicators distinguish short wetting events from repeated or prolonged exposure, identify delayed drying, and communicate data reliability. The Peavy Hall dataset reinforced this logic: at one floor connection, upper and bottom plies reached similar peak MC values of about 44–45 % but returned to 16 % MC after approximately 25 and 92 days, respectively (Baas et al., 2021).

A lifecycle-oriented monitoring record should therefore do more than store time-series data. It should preserve the contextual and derived information needed to interpret moisture behaviour over time, including repeated wetting, prolonged high-moisture periods, delayed drying, unexplained anomalies, and data quality. These interpreted indicators provide a more usable basis for maintenance planning, post-event assessment, durability evaluation, and future reuse-related assessment than isolated sensor readings.

3.4 Decision support and lifecycle integration — Layers 7–8

The output and use component covers layers 7 and 8 of the framework, decision support and lifecycle data integration. These layers describe how interpreted moisture exposure histories are connected to actions during construction, operation, and future assessment.

During construction, the record may support inspection priorities, drying verification, or decisions before enclosure. In operation, it may support leak detection, maintenance planning, and mould- or service-life-related assessment (Flexeder et al., 2022; Schmidt & Riggio, 2019). The framework extends this evidence by identifying additional uses, such as targeted opening of concealed assemblies and post-event assessment following incidents (e.g. plumbing, roof, façade, or flooding). At end of life, the same record may provide supporting evidence on whether an element experienced isolated wetting with verified drying, repeated wetting, prolonged high-moisture periods, or uncertain data quality.

Decision support is based on interpreted indicators rather than isolated sensor readings. In the Peavy Hall dataset, local detail behaviour differed substantially from nearby reference behaviour. A lower-shearwall middle-panel location remained below 16% MC, while an edge-by-hardware location reached about 27.6% MC and remained above 20% MC for about 88 days (Baas et al., 2021). This illustrates the value of comparing high-risk locations with baseline or reference locations, since such comparisons help distinguish local wetting or delayed drying from broader seasonal or indoor-climate behaviour. In the framework, such comparisons support inspection, maintenance, post-event

assessment, and reuse-related evaluation, while keeping moisture exposure history as supporting evidence rather than a standalone decision criterion.

The literature showed limited direct guidance on long-term information management for moisture-monitoring records. The final layer therefore concerns lifecycle data integration with digital tools, where interpreted monitoring records are linked to information needed for later use, including element identity, sensor metadata, exposure indicators, data-quality flags, and decision history. Digital building logbooks, product passports, and related lifecycle information systems are relevant because they are intended to retain building- and product-level information across lifecycle stages (Tang et al., 2025). BIM objects, digital twins, monitoring platforms, and avatar-assisted SHM systems can also support transfer and reuse of lifecycle moisture records by linking monitoring data to spatial, material, and use-phase context (Bucher et al., 2024).

A practical lifecycle record links element and location data, sensor configuration, risk class, validated exposure indicators, data-quality flags, and maintenance, inspection or repair history, making moisture monitoring part of a traceable lifecycle information process for future reuse.

4. DISCUSSION

The framework makes the reasoning behind moisture monitoring more explicit. It positions monitoring as an engineering judgement shaped by moisture risk, monitoring objective, and later interpretation needs. Existing timber-monitoring studies provide valuable evidence on vulnerable locations, sensor technologies, installation challenges, and data-acquisition methods (Flexeder et al., 2022; Johns et al., 2024; Schmidt & Riggio, 2019; Uwizeyimana et al., 2020). However, the translation from location-specific moisture risk to monitoring strategy often remains implicit. By making this translation visible, the framework provides a way to document why particular locations, depths, levels of redundancy, and protection measures are selected.

This framing also clarifies what kind of guidance the framework provides. It does not propose universal sensor spacing, fixed sensor densities, or standardised layouts because moisture risk is not evenly distributed across a building. The usefulness of a sensor measurement depends on the monitored detail, likely moisture mechanism, access after enclosure, drying capacity, measurement depth, and intended decision use. The framework therefore supports consistent reasoning while allowing project-specific judgement. Practical implementation will also depend on sensor, installation, connectivity, maintenance, and replacement costs; these constraints should be examined in future case studies.

The emphasis on contextualised records is important because moisture measurements are only useful for durability assessment when they can be interpreted over time and in relation to their local conditions. The aim is not simply to detect that a moisture threshold was crossed, but to preserve enough information to judge the severity and reliability of the exposure history. Studies on moisture-related degradation in timber show that duration, recurrence, drying behaviour, temperature conditions, and local assembly conditions influence durability risk, not peak MC alone (Bornemann et al., 2014; Brischke & Rapp, 2008; Shirmohammadi et al., 2021; Viitanen, 1996). The analysed monitoring datasets reinforced this point by showing that interpretation depends on localised exposure, measurement depth, drying time, and data quality. This supports the framework's emphasis on contextualised records rather than isolated values.

The lifecycle information aspect of the framework should be understood as an interpretation problem as well as a storage problem. From a reuse perspective, a lifecycle record can reduce uncertainty about an element's condition and exposure history at recovery, supporting later assessment for reuse. Material-passport approaches illustrate how structured component information can be retained for later material assessment (Honic et al., 2019).

Digital building logbooks, product passports, and BIM objects can retain element-level material, spatial, maintenance, inspection, and intervention information, while digital twins and monitoring platforms can support data access, visualisation, validation, and interpretation during use (Tang et al., 2025). Moisture monitoring can complement these systems where readings remain linked to sensor depth, risk class, data-quality flags, exposure indicators, and relevant interventions. Lifecycle integration therefore requires both a digital carrier and sufficient contextual information to preserve the meaning of the monitoring record for later circular assessment.

The framework remains conceptual and should be read as structured guidance rather than a validated decision rule. Its risk classes, deployment responses, and information requirements represent a synthesis of current evidence and should not yet be treated as formal design or reuse-assessment criteria.

5. CONCLUSION

This paper presents a lifecycle information framework for moisture monitoring in mass-timber buildings. The main contribution is the integration of moisture-risk identification, risk-based sensor deployment, monitoring metadata, data-quality validation, and moisture-exposure evaluation into a single planning structure. In doing so, the framework shifts moisture monitoring from isolated sensor measurement towards the production of interpretable records that can support maintenance planning, post-event assessment, durability evaluation, and future reuse-related decision-making.

The framework shows how moisture-critical zones can be structured across building, assembly/detail, and element/material-feature scales. These locations can then be classified using criteria such as exposure likelihood, drying capacity, inspection access, consequence, environmental aggressiveness, sensor survivability, and decision relevance. This classification provides a basis for risk-informed deployment decisions, including sensor placement, density, redundancy, measurement depth, reference locations, and protection requirements.

The framework also identifies core information needed for later interpretation of monitoring records. In addition to measured values such as MC, RH, temperature, timestamps, sensor identity, and measurement depth, records require contextual information on element identity, sensor location, risk class, installation context, calibration or correction basis, data-quality flags, exposure indicators, and relevant maintenance or intervention history. With this information, moisture readings can be interpreted as exposure histories rather than isolated values.

For reuse assessment, these histories can reduce uncertainty about whether an element experienced isolated wetting with verified drying, repeated wetting, prolonged elevated moisture, or uncertain monitoring conditions. They should support, rather than replace, broader assessment of structural capacity, connections, visible damage, biological degradation, deconstruction damage, and regulatory requirements.

Future research should test the framework in monitored mass-timber projects and practitioner review, examining whether its risk-classification logic, deployment responses, and information requirements can be applied consistently during construction and operation. It should also assess whether contextualised moisture-exposure histories, combined with evidence on structural condition and deconstruction damage, improve maintenance, post-event assessment, and reuse evaluation. Further work is needed on practical data templates, semantic structures, and interpretation criteria for retaining usable records across the building lifecycle.

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